

Brac University
Department of Electrical & Electronic Engineering
Spring2026

Course Number : EEE308L

Section :06

Group No :04



Lab Report

Experiment no: 04

Name of the experiment:

Study of Operational Amplifier as Amplifier, Comparator and Precision Rectifier using PSpice

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Objective:-

The objective of this experiment is to understand the operation of different Op-Amp circuits by designing and simulating them. These include non-inverting amplifiers, inverting amplifiers, voltage follower, and precision rectifier. The aim is to observe voltage gain, input-output relationship, and waveform behavior using simulation tools.

Equipment:-

- Operational Amplifier (Op-Amp)(uA741)
- Resistors
- Function Generator (for input signals like sine and pulse)
- Power Supply
- Graph Viewer
- Simulation Software (Pspice)
- Connecting wires

Lab Tasks:-

1. Design an amplifier using an Op-Amp that produces a gain of 10V/V.

Answer:-

Circuit Diagram:

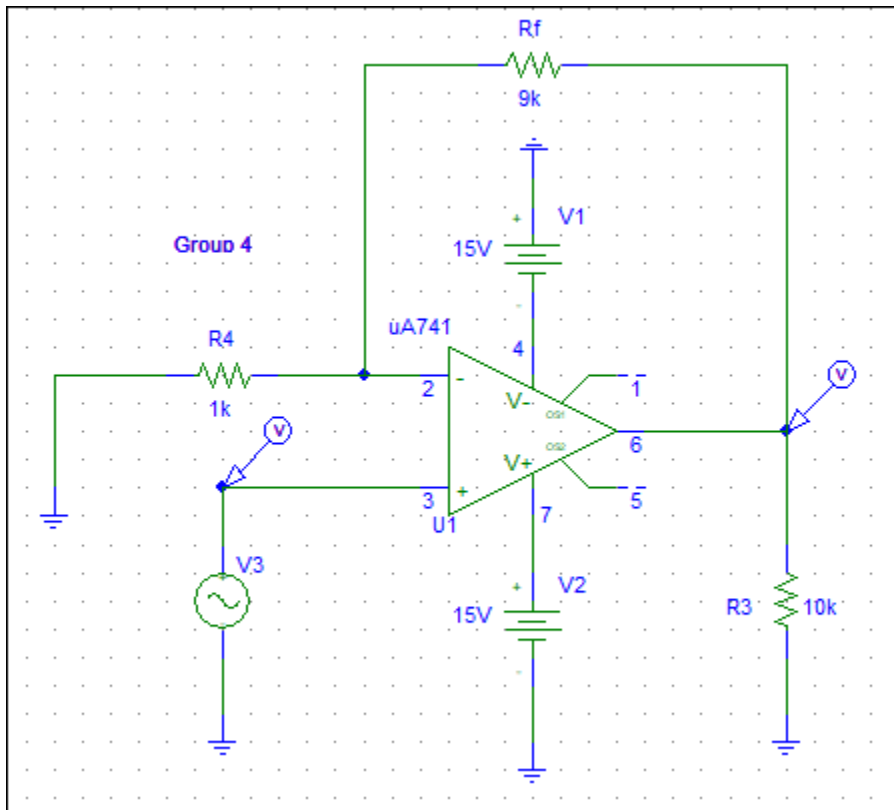


Figure : Non-Inverting Operational Amplifier for a closed-loop voltage gain of 10V/V.

Transient Parameter

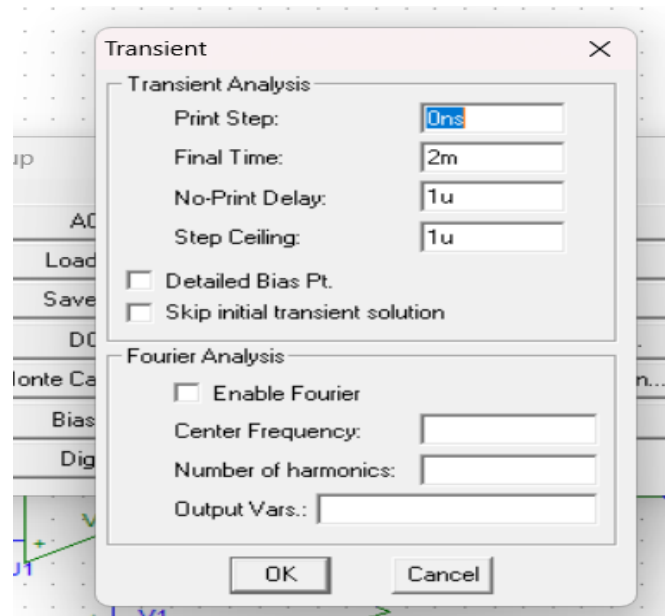


Figure : Transient Analysis Setup for Non-Inverting Amplifier

V_i & V_o Graph:

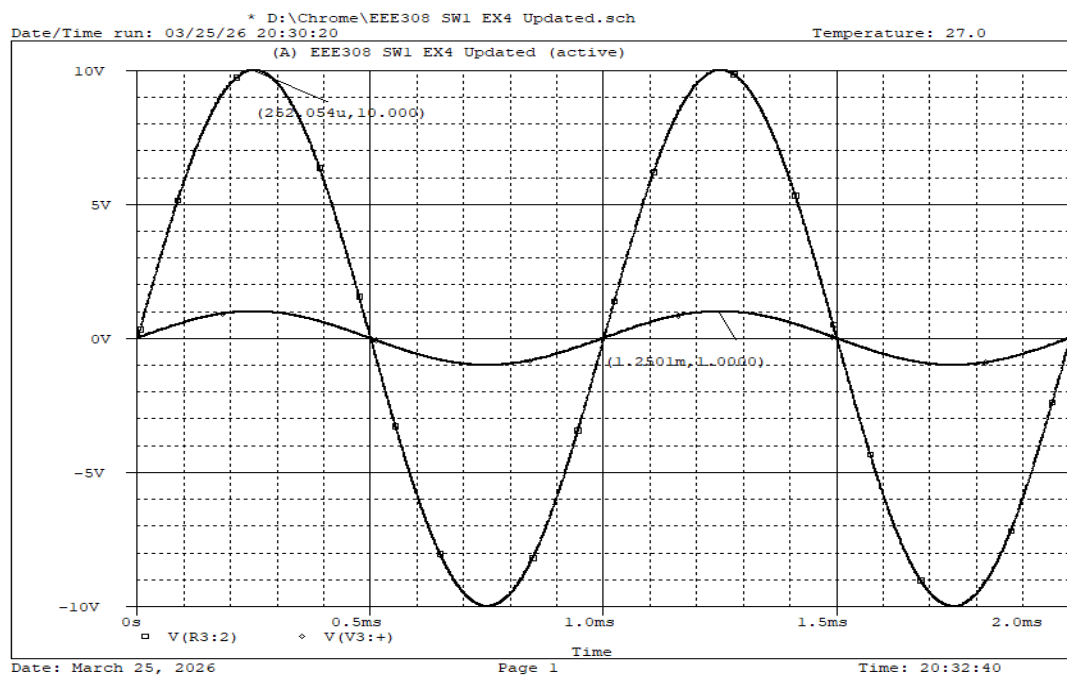


Figure : Input (V_i) and Output (V_o) Waveform of Non-Inverting Amplifier With 10v gain

Calculation:-

We know,

Voltage gain for Non-Inverting Operational Amplifier, $A_v = 1 + \frac{R_f}{R_i}$

Here,

$$R_f = 9\text{k}\Omega$$

$$R_i = 1\text{k}\Omega$$

$$\begin{aligned} A_v &= 1 + \frac{R_f}{R_i} \\ &= 1 + \frac{9}{1} \\ &= 10 \text{ V/V.} \end{aligned}$$

2. Simulate an inverter and a voltage follower using an Op-Amp.

Answer:-

Circuit Diagram:

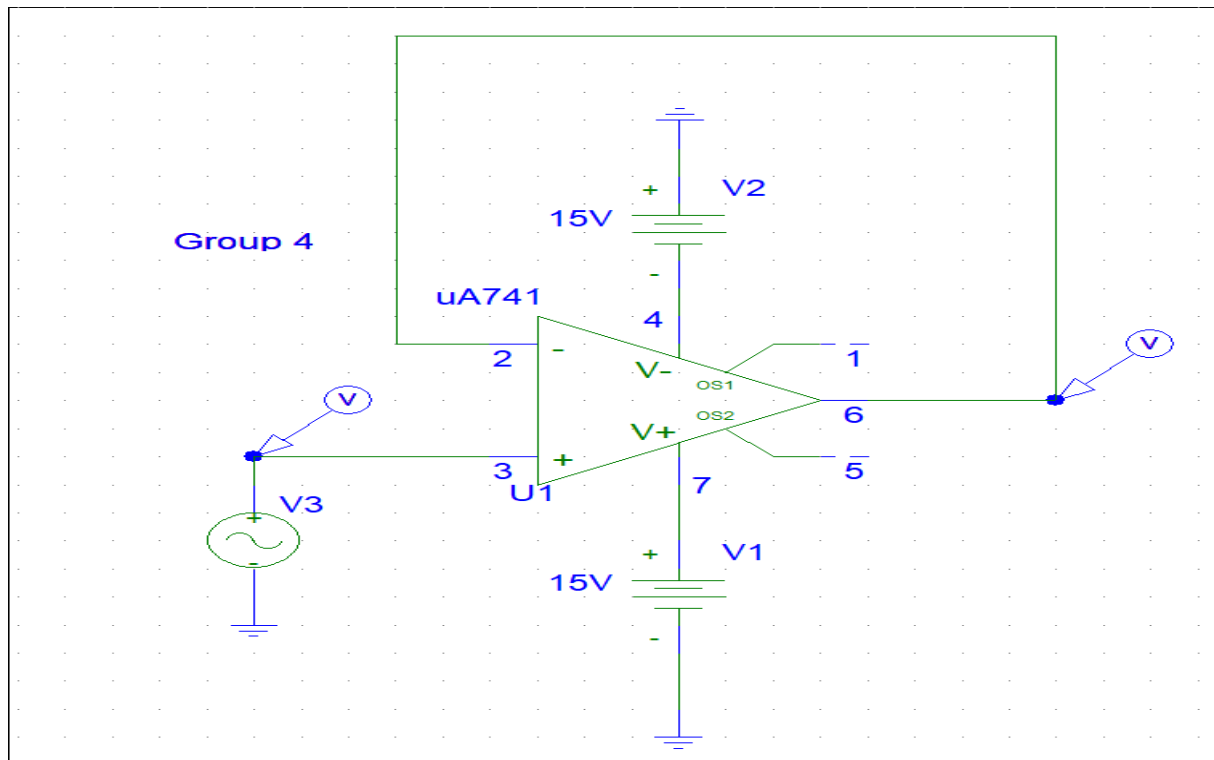


Figure : voltage follower using an Op-Amp

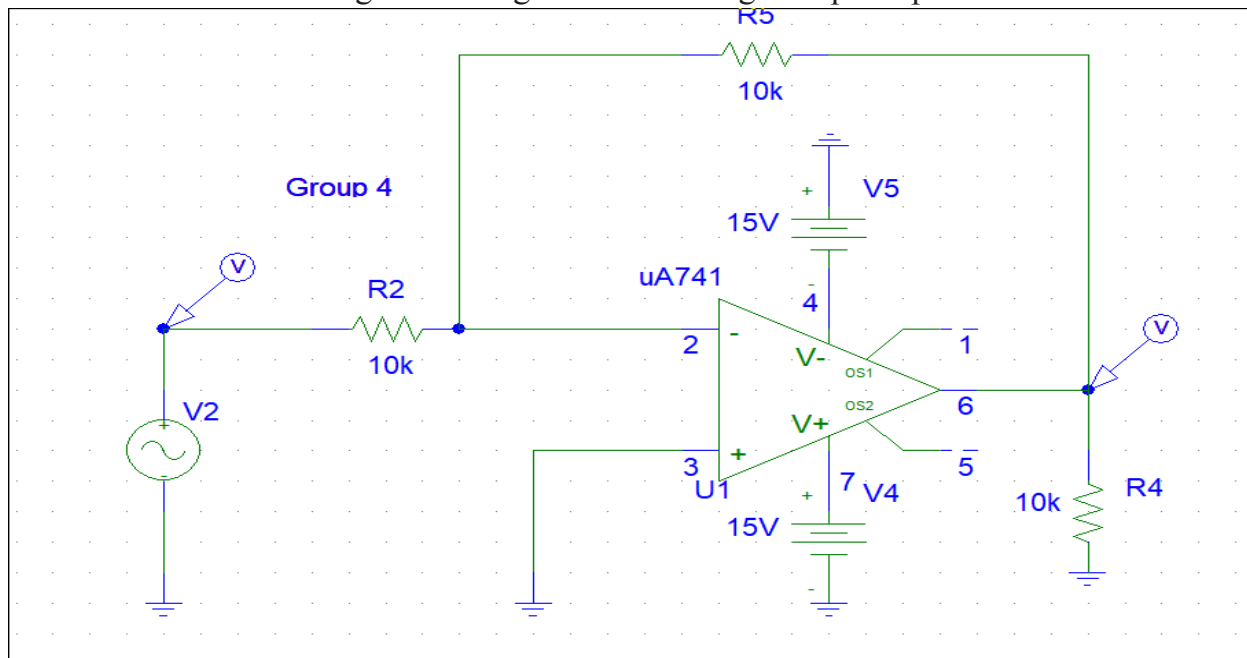


Figure : Inverter using an Op-Amp

Transient Parameter

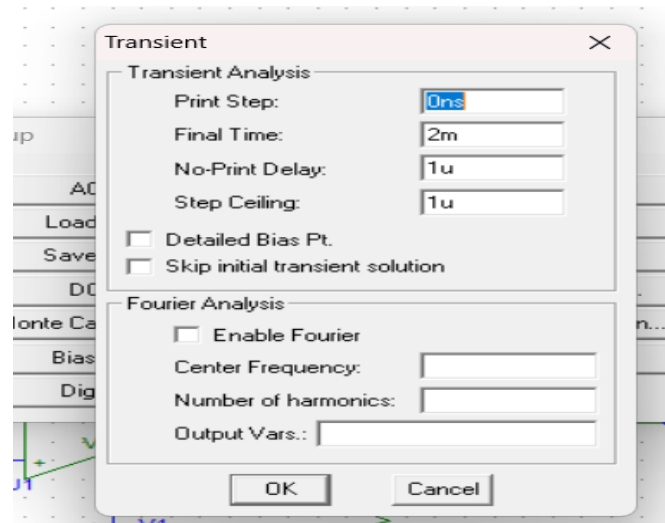


Figure :Transient Analysis Setup for Inverter and Follower
 V_i & V_o Graph:

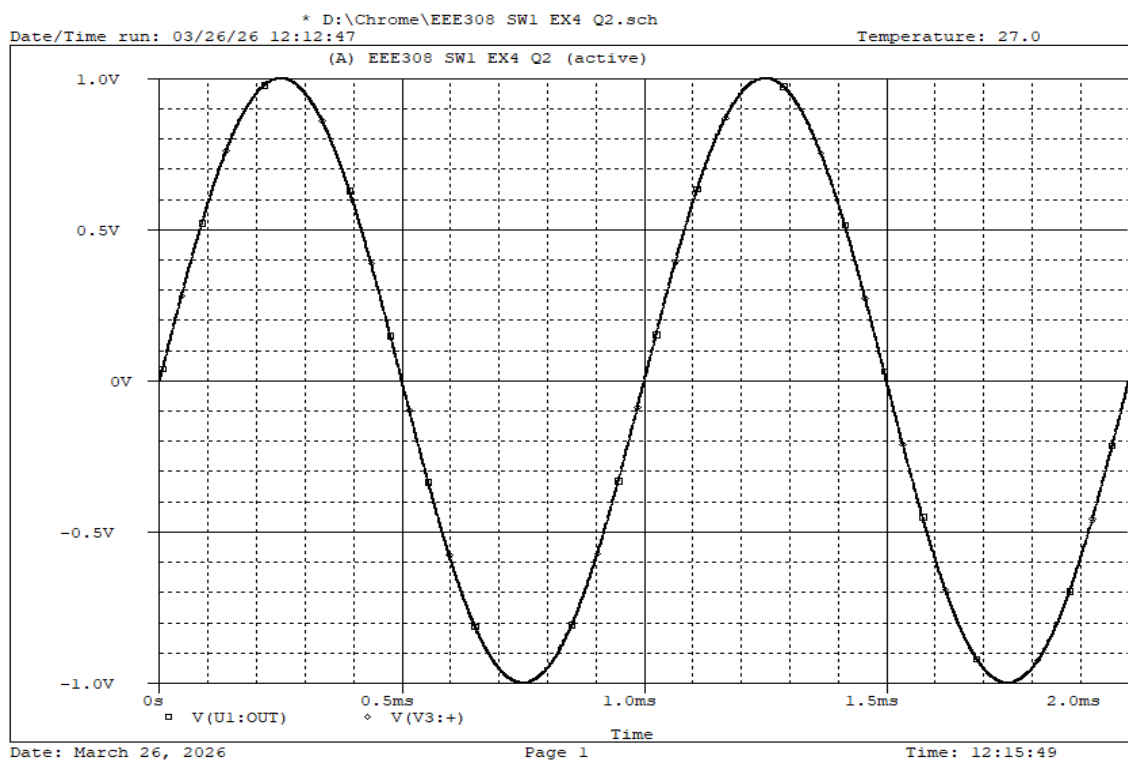


Figure :Input (V_i) and Output (V_o) Waveforms for voltage follower($V_i=V_o$)

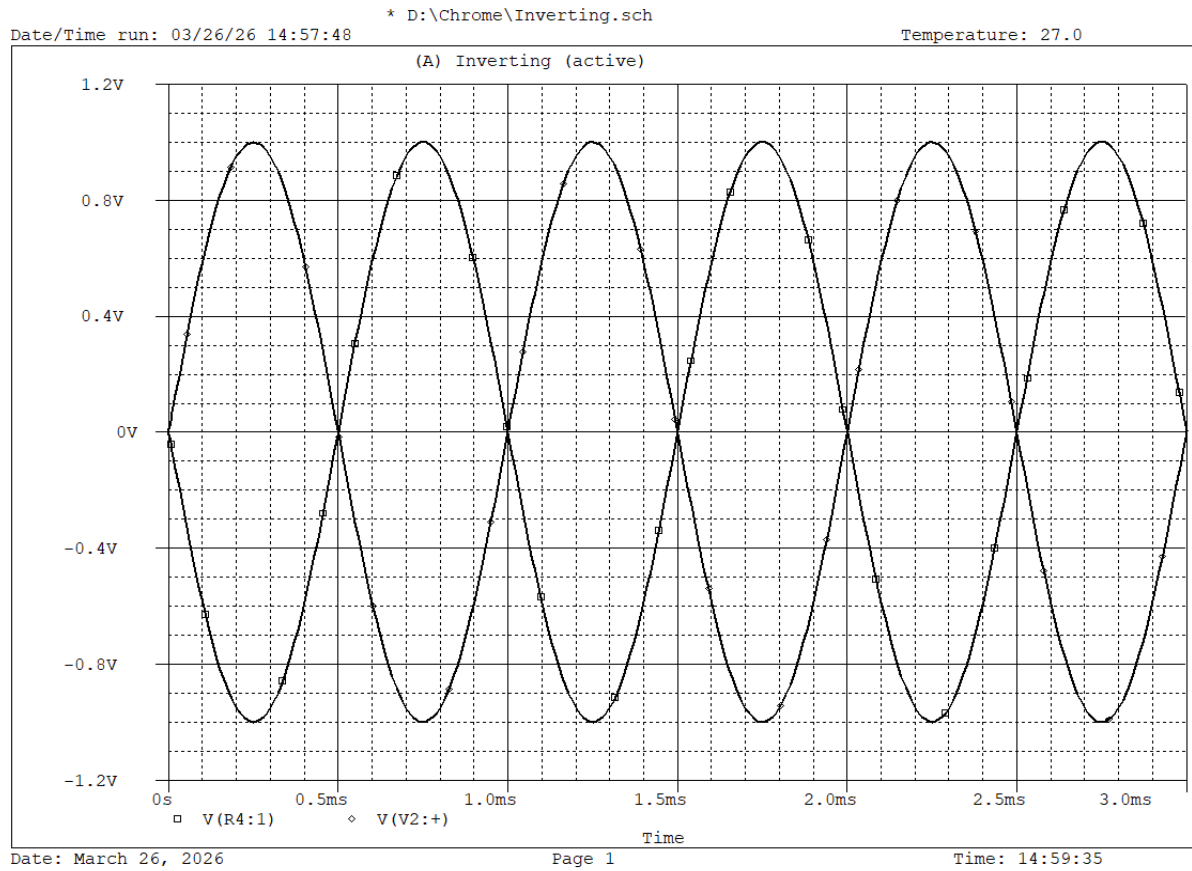


Figure :Input (V_i) and Output (V_o) Waveforms for Inverter

3. Design an amplifier using an Op-Amp with gain of 0.3V/V.

Answer:-

Circuit Diagram:

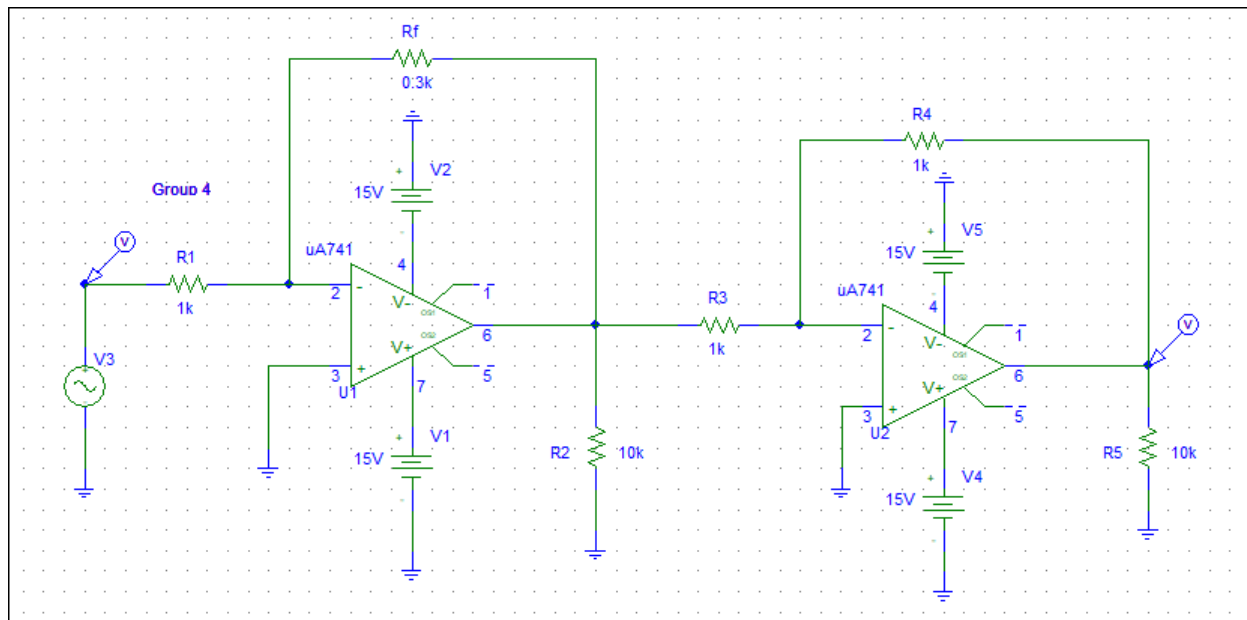


Figure : Two-Stage Op-Amp Circuit for Gain = 0.3 V/V

Transient Parameter

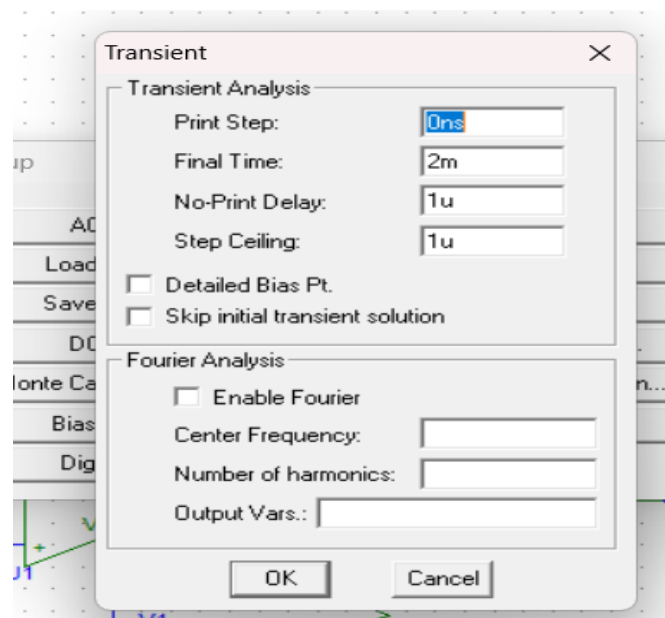
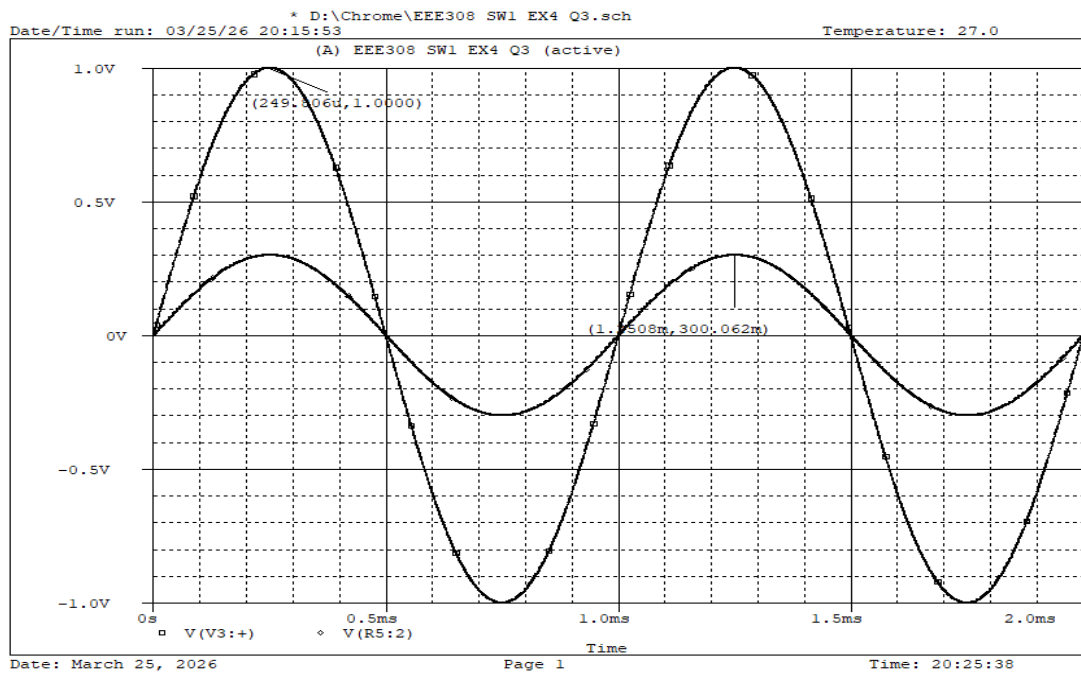


Figure : Transient Analysis Setup

V_i & V_o Graph:



Calculation:-

- Stage 1 (Left): This is an inverting amplifier. The gain is

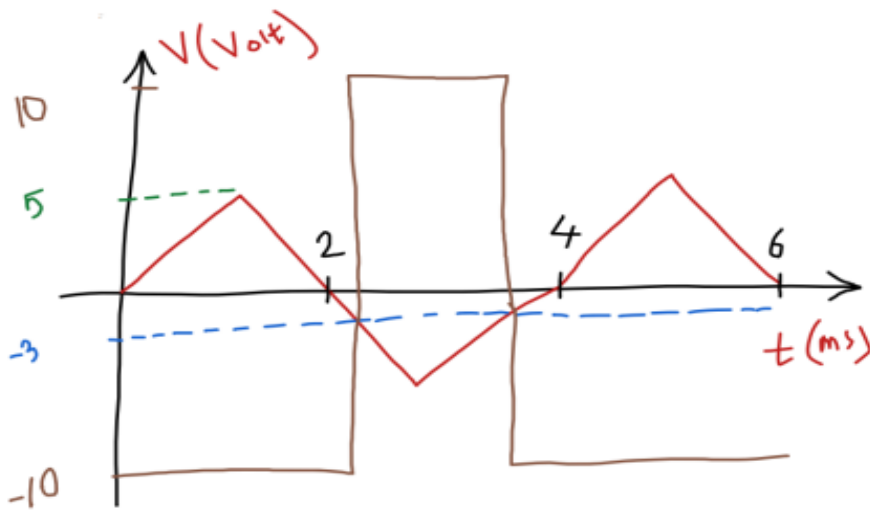
$$A_1 = - \frac{R_f}{R_1} = - \frac{0.3k}{1k} = - 0.3$$

- Stage 2 (Right): This is another inverting amplifier. The gain is

$$A_2 = - \frac{R_3}{R_4} = - \frac{1k}{1k} = - 1$$

$$\begin{aligned} \text{Total gain } A_{\text{Total}} &= A_1 * A_2 \\ &= (-0.3) * (-1) \\ &= 0.3 \text{ V/V.} \end{aligned}$$

4. Design a circuit that follows the following input-output relationship:



Answer:-

Circuit Diagram:

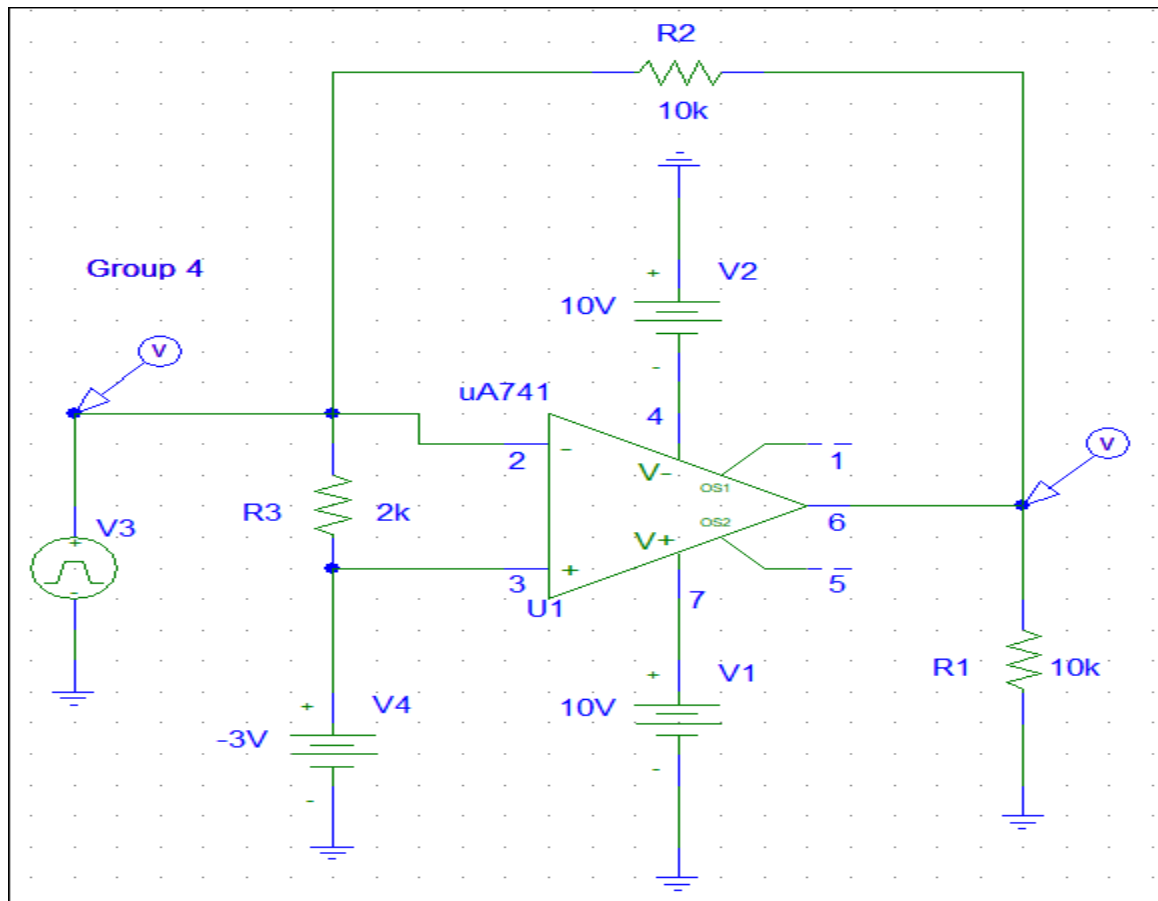


Figure :Designed Circuit for Given Input-Output Relationship

Transient Parameter

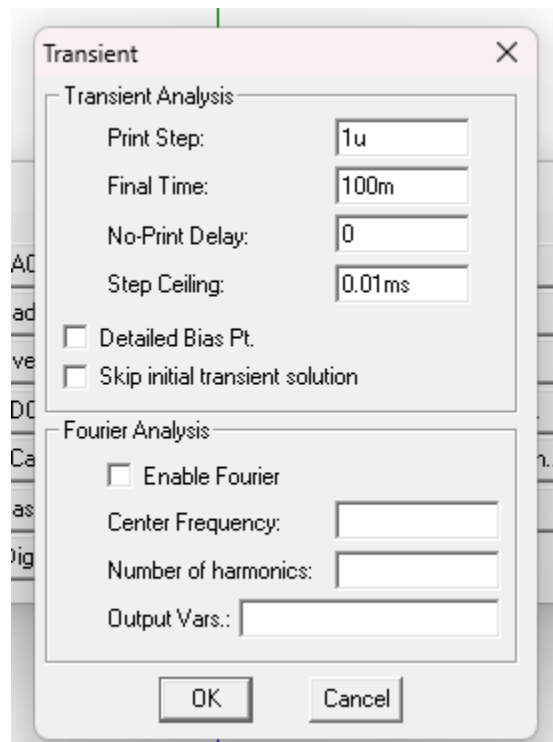


Figure : Transient Analysis Setup

V_{Pulse} Parameter:-

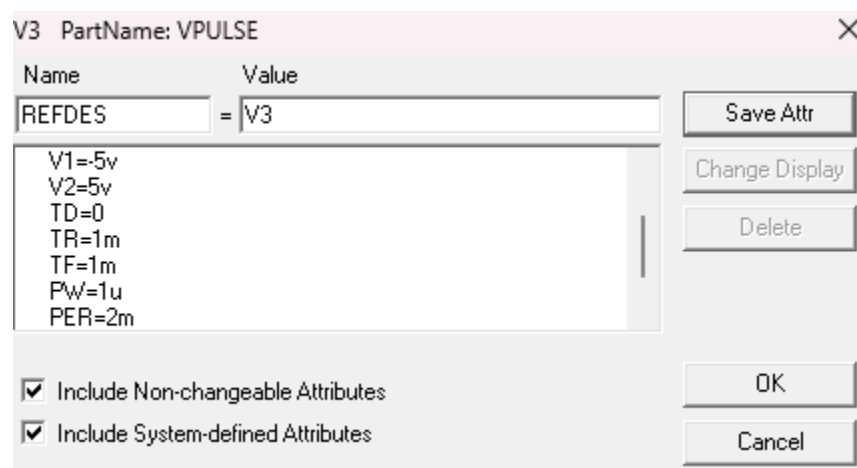


Figure :VPulse Input Parameter Settings

V_i & V_o Graph:

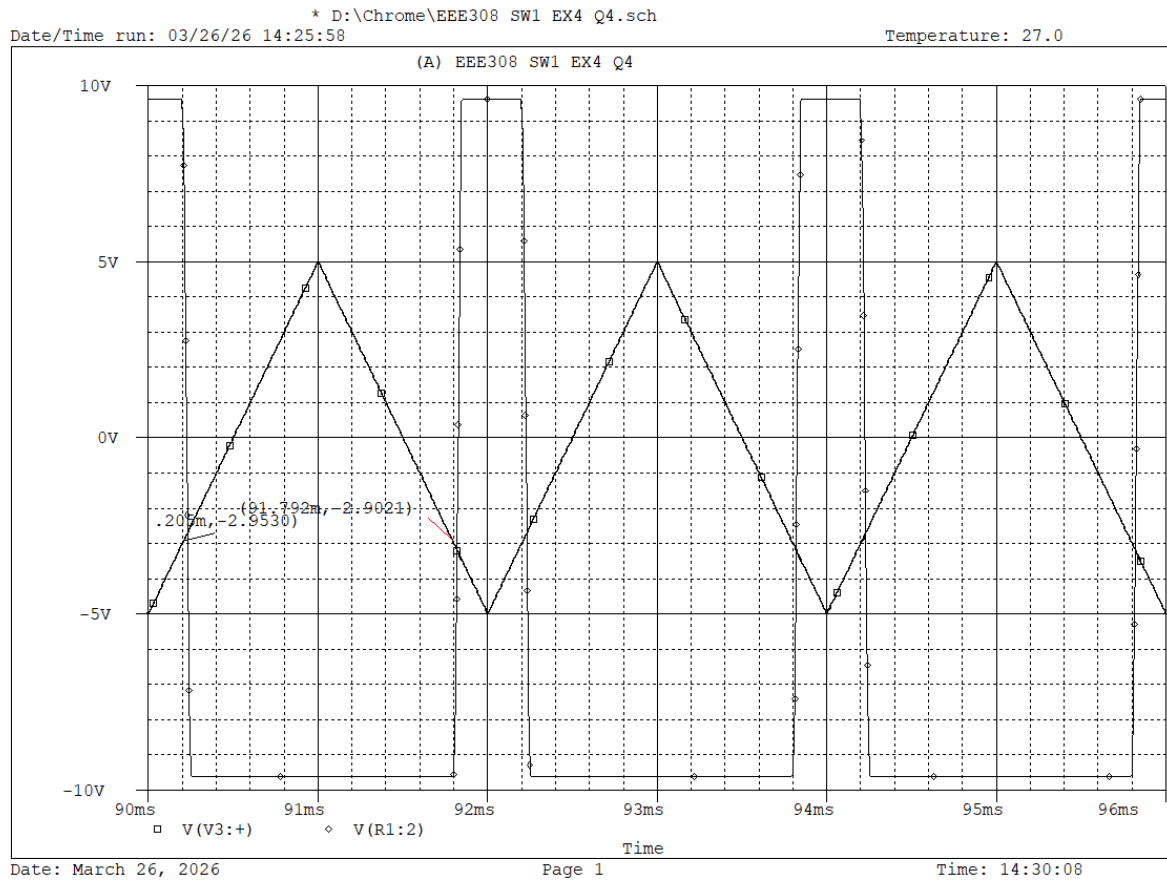


Figure : Input (V_i) and output (V_o) waveforms demonstrating how the circuit responds to pulse input and satisfies the required input-output relationship.

5. Show the output of the precision rectifier if the input voltage has a peak amplitude of 10mV.

Answer:-

Circuit Diagram:-

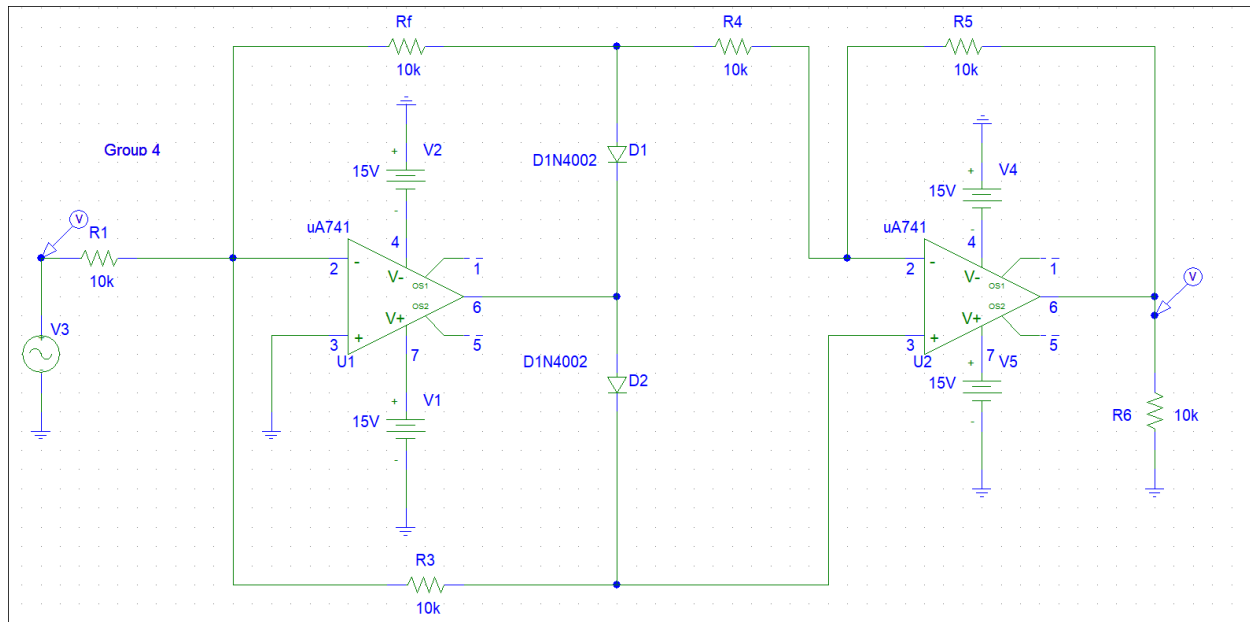
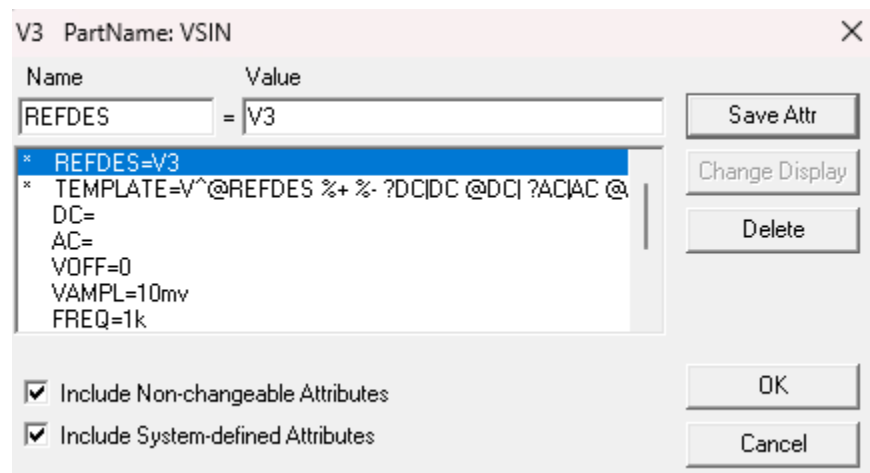


Figure :Circuit diagram of a precision rectifier

V_{sin} parameter:-



Transient Parameter

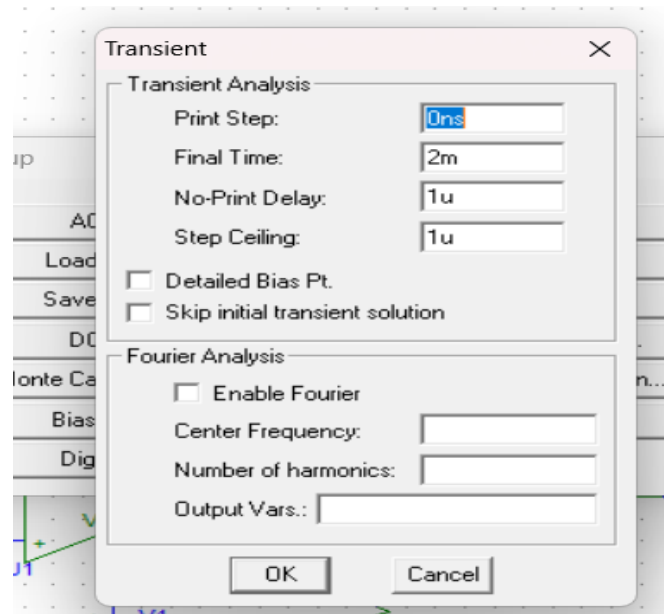


Figure : Transient Analysis Setup

V_i & V_o Graph:

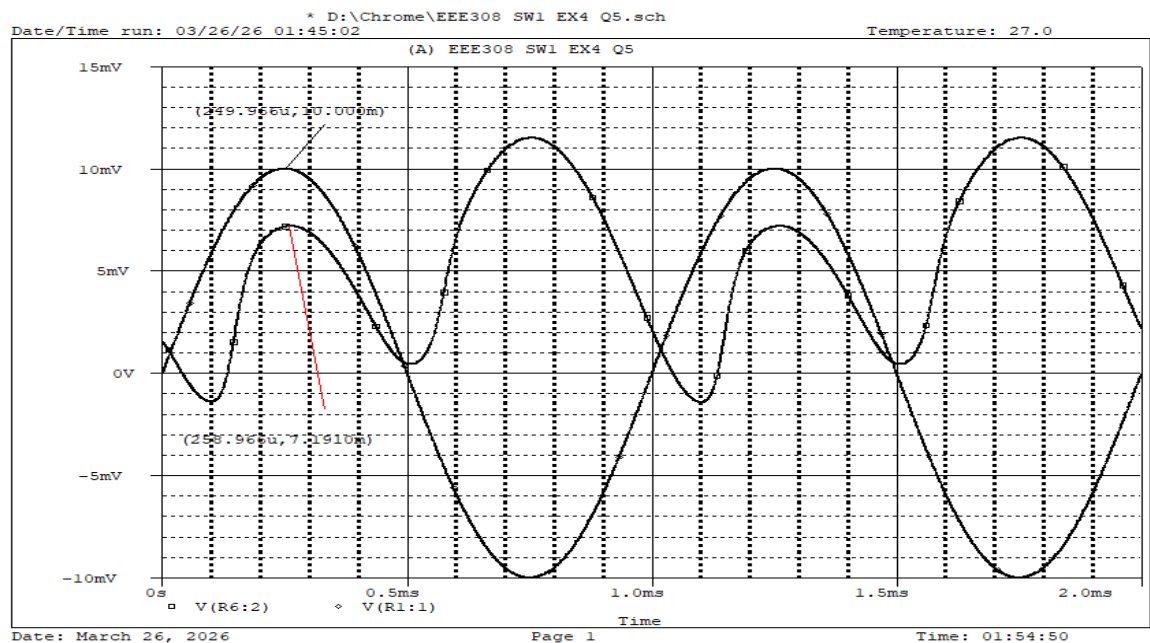


Figure : Input and output waveforms illustrating precise rectification of the small input signal, where the output follows only the positive half-cycle with high accuracy.

Conclusion:-

In this experiment, we designed and simulated different Op-Amp circuits such as non-inverting amplifiers, inverting amplifiers, voltage follower, and precision rectifier. From the results, we saw that the output matched the expected gain and behavior of each circuit. The non-inverting amplifier gave a gain of 10, while the two-stage amplifier gave a smaller gain of 0.3. The inverting amplifier showed phase inversion, and the voltage follower gave the same output as input. We also observed how the circuits respond to different input signals using transient analysis. The precision rectifier worked properly even for very small input voltage, which shows its advantage over normal diode circuits. Overall, this experiment helped us clearly understand how Op-Amps work in different configurations and how they are used to control and modify signals in practical applications.